

CarMate

An AI-Powered SDV Blueprint for Intelligent In-Vehicle
Companions

Chris Cheng – Ferdinand-Steinbeis-Institut



 www.transportbranche.de

Truck drivers on the road: Lonely nights and the desire for Company

Life on the road is demanding – long journeys, tight schedules, and constant distance from family and friends are part of everyday life for Professional Drivers.



 www.ruhrnachrichten.de

Lothar (61) has been driving trucks for 40 years. He talks about loneliness and romance at the truck stop.

Lothar Westbrock from Dortmund wanted to be a truck driver even as a teenager. After 40 years, he knows the romantic side of the profession – but also its downsides.



 www.augsburger-allgemeine.de

Series of attacks on the B2: Truck driver threw stones out of boredom

A 49-year-old man caused fear and panic on a federal highway near Augsburg by throwing stones. In court, he admitted to the charges but remained guilty of one offense.



 www.wp.de

Truck Drivers on the A45 - Loneliness on the Road

Truck drivers cover long distances on highways and spend days and nights alone in their vehicles. A life (almost) without social contact.

Solution

Why "CarMate" Blueprint is Needed:

Driver loneliness problem - Long-distance drivers face isolation and mental stress, which can impact safety and well-being

Need for intelligent in-car interaction - Drivers need a natural, voice-based assistant to control vehicle features and get real-time info

The Goal of this Blueprint

Better human-vehicle connection - Turns the vehicle into a "companion" instead of just a machine, improving user experience

- Design and implement CarMate, an AI-based vehicle assistant integrated via the Eclipse SDV ecosystem and digital.auto Playground, enabling standardized communication and interaction across distributed SDV components.



LLM-based AI-Agent



MQTT



paho

Eclipse SDV

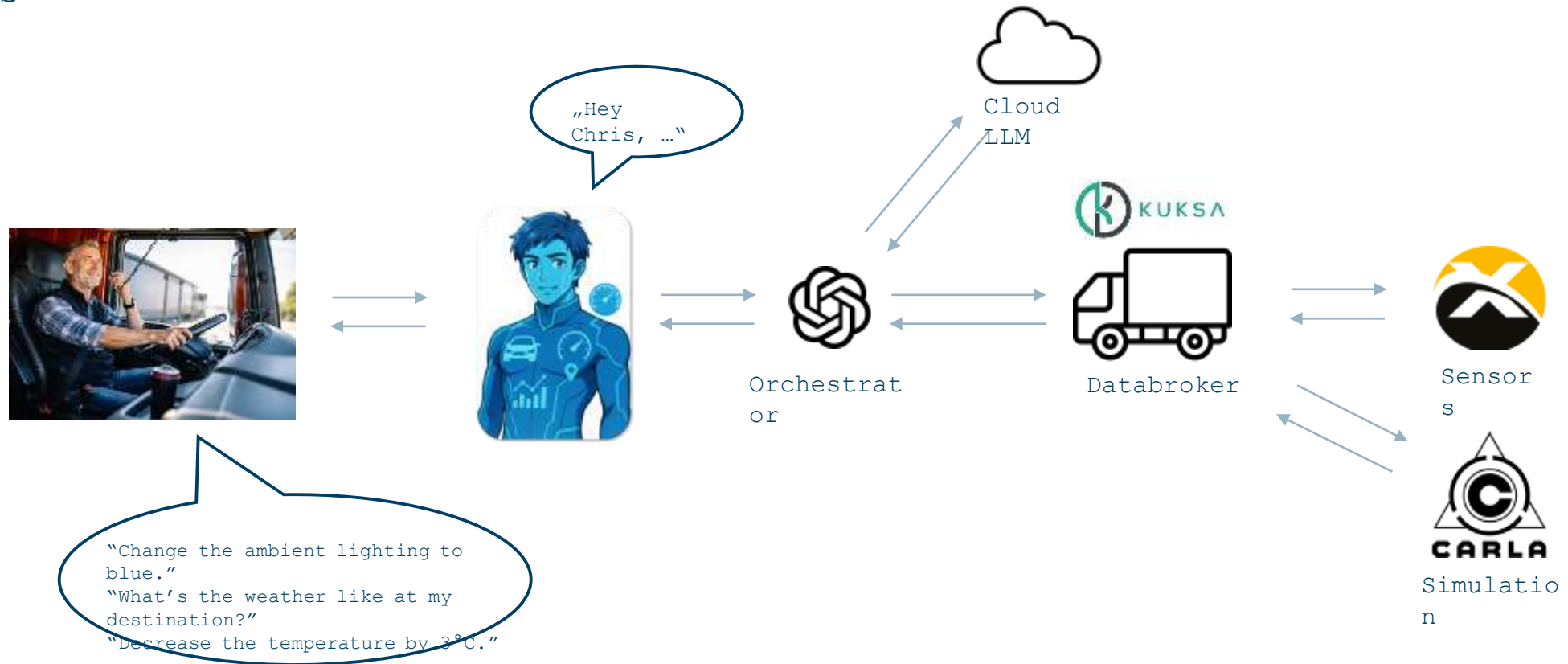


KUKSA



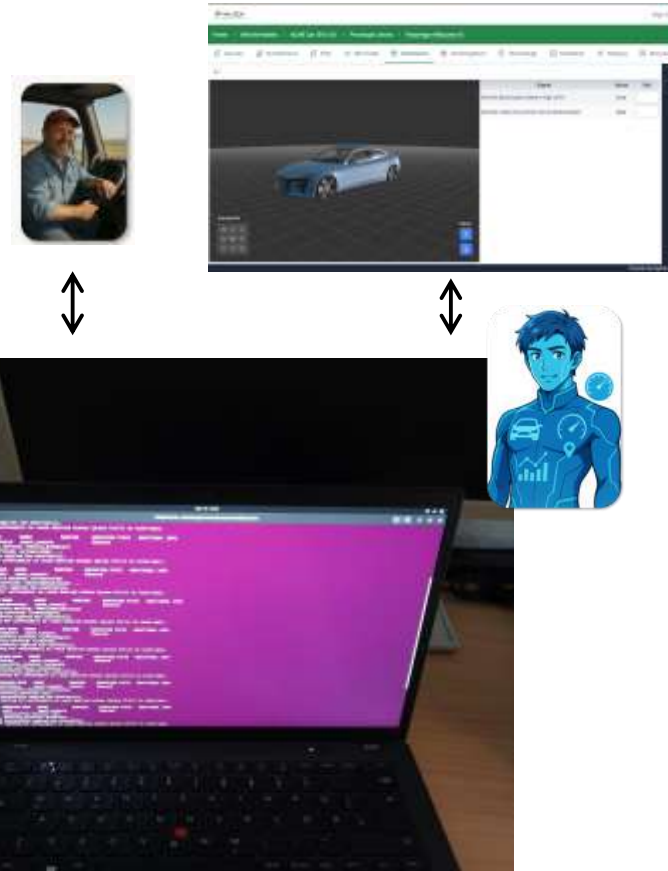
CarMate

-- Your Smart Companion for Long Journeys





MCU Node

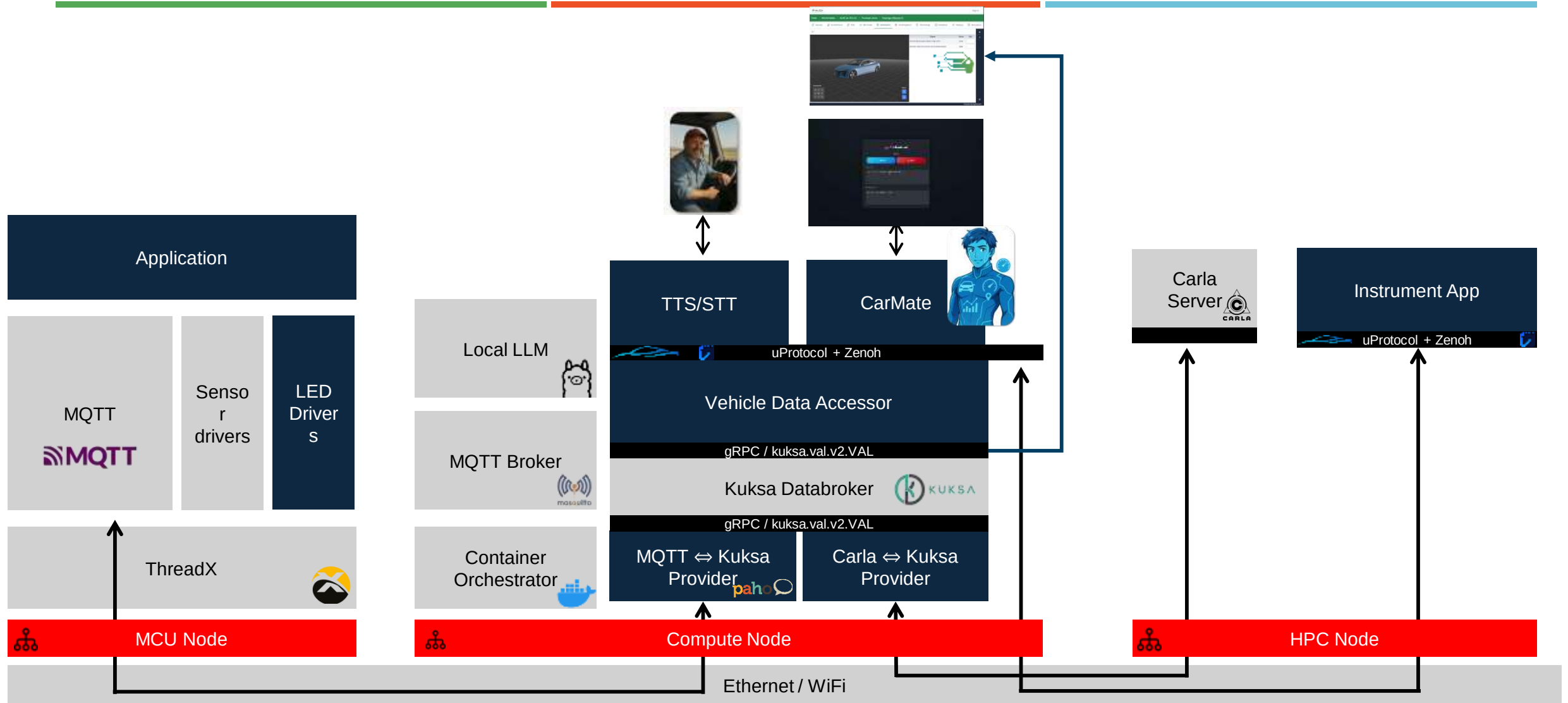


Compute Node



HPC Node

Technical Architecture



- **Docker** - workload orchestration in SDV environment
- **MQTT + Eclipse Paho** - communication between MCU and compute nodes
- **KUKSA Databroker** - central vehicle data management (VSS model)
- **uProtocol + Zenoh** - communication layer between services
- **CARLA Simulator** - virtual vehicle + environment simulation
- **gRPC (kuksa.val.v2)** - data access interface
- **ThreadX (RTOS)** - MCU-level embedded system
- **Rust & Python** - core development languages
- **TTS / STT modules** - voice interaction system
- **LLM integration** - conversational AI assistant (CarMate brain)

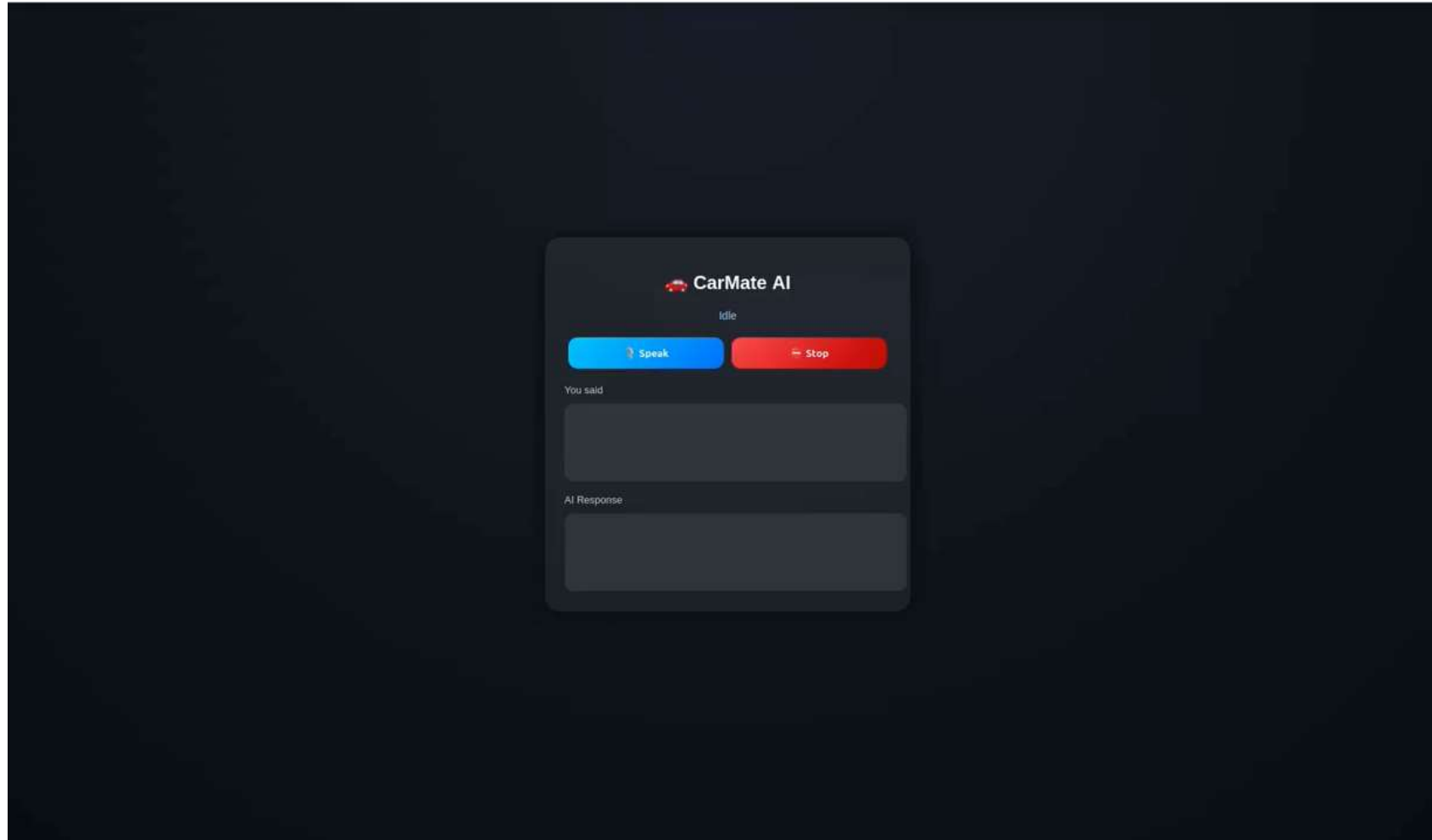
Hardware setup

Simulation Environment: The CARLA simulator is hosted on the secondary (left) laptop.

Application Hosting: An Ethernet connection links the secondary hardware to the primary (right) laptop, which runs the dedicated CarMate application.

Hardware Integration: An integrated Microcontroller Unit (MCU) facilitates the transmission of real-time temperature data and manages the control of ambient lighting systems.





Key Contribution:

- An SDV blueprint demonstrating how Eclipse open-source tools enable AI-driven in-vehicle companion systems
- Integration of Eclipse SDV tools (KUKSA, uProtocol, Zenoh, Autowrx, etc.)
- Standardized communication across vehicle domains
- Bridging cloud, edge, and embedded systems

Future Work:

- Fine-Tune LLM to understand driver's emotion.
- Control cruise control with voice commands.
- Integrate Camera for Gesture Control and Driver's emotions recognition.
- Integrate the project with EDCAR.

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